

Typed Period-Prime Carrier Selectors

A Finite Selection Theorem for the Triple $\Omega = (137, 107, 103)$

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Abstract

This paper formalizes a finite selector mechanism on typed prime pairs (w, q) , where w is a prime channel value and q is an odd prime divisor of $w - 1$. The mechanism, called the Typed Period-Prime Carrier Selector (TPCS), specifies three channels α , SU2, PC2, each defined by Legendre-symbol orientation gates and 2-adic lane conditions on $w - 1$. The displayed output of each channel is the w -projection of the selected typed pair.

The main result is a finite minimality theorem. Under the canonical strong joint-tile TPCS rules, with joint transport boundary

$$M = 80 = 2^4 \cdot 5 = \text{lcm}(16, 5),$$

the first channel-wise typed carrier pairs are

$$\alpha = (137, 17), \quad \text{SU2} = (107, 53), \quad \text{PC2} = (103, 17).$$

Projecting to w -coordinates gives $\Omega = (137, 107, 103)$.

The proof is finite. It verifies that the target pairs pass all gates and that no earlier admissible pair exists in the only intervals required for minimality:

$$80 \leq w < 137, \quad 80 \leq w < 107, \quad 80 \leq w < 103.$$

The theorem is independently crosschecked by two proof kernels and hardened by proof-object mutation testing.

The paper does not claim that TPCS is the unique selector grammar. It proves that the stated selector has a finite, verifiable minimum theorem and that its surrounding mechanism has causal structure, reset behavior, counterfactual teeth, and bounded adversarial robustness.

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1 Introduction

Companion papers [9, 10] establish a local and projective arithmetic substrate for finite power tables: row-column transport, gcd-depth, quotient-unit structure, mirror signatures, anchored CRT position, and projective compatibility under divisor chains.

This paper moves from transport to selection. It studies a finite typed selector mechanism acting on period-prime carriers. The selector does not operate on isolated primes. It operates on pairs (w, q) , where w is the displayed channel value and q is a prime carrier inside the multiplicative structure of $w - 1$.

The central claim is deliberately limited:

Given the canonical TPCS strong joint-tile rules, the first typed carrier pairs are $(137, 17)$, $(107, 53)$, $(103, 17)$, and their w -projection is $\Omega = (137, 107, 103)$.

This is a finite theorem relative to a stated mechanism. It is not a claim that all possible grammars must produce Ω .

2 Typed period-prime carriers

Definition 1 (Typed period-prime carrier). A *typed period-prime carrier* is a pair (w, q) where:

1. w is prime,
2. q is an odd prime,
3. $q \mid (w - 1)$.

The value w is the *displayed channel coordinate*. The value q is a *period-prime carrier* certifying structure inside $w - 1$.

The selector output is not the raw pair. It is the projection

$$\pi_w(w, q) = w.$$

The final displayed triple $\Omega = (137, 107, 103)$ is therefore the w -projection of the typed carrier triple $((137, 17), (107, 53), (103, 17))$.

This distinction is essential. Earlier formulations using the full odd part of $w - 1$ as the carrier fail for the third channel: $103 - 1 = 102 = 2 \cdot 3 \cdot 17$ has odd part 51, while the correct period-prime carrier is $q = 17$. TPCS therefore uses prime-divisor carriers, not global odd-part carriers.

3 Shared carrier conditions

A typed carrier (w, q) is required to satisfy the shared period-carrier gates:

$$\begin{aligned} w &\text{ prime,} \\ q &\text{ odd prime,} \\ q &\mid (w - 1), \\ q &\equiv 1 \pmod{4}, \\ q &> \sqrt{w}. \end{aligned}$$

The condition $q \equiv 1 \pmod{4}$ is an orientation condition on the period carrier. The condition $q > \sqrt{w}$ makes q a large carrier of the period envelope rather than a small divisor. As shown later (Section 14), these are not always first-output critical, but they are structural-hygiene gates: removing them admits off-mechanism rows while sometimes leaving the first selected triple unchanged.

4 Channel types

The selector has three channel types: α , SU2, and PC2.

α channel

A typed carrier (w, q) is α -admissible if it satisfies the shared carrier conditions and

$$v_2(w - 1) = 3, \quad \left(\frac{2}{q}\right) = 1.$$

SU2 channel

A typed carrier (w, q) is SU2-admissible if it satisfies the shared carrier conditions and

$$v_2(w - 1) = 1, \quad \left(\frac{2}{q}\right) = -1.$$

PC2 channel

A typed carrier (w, q) is PC2-admissible if it satisfies the shared carrier conditions and

$$v_2(w - 1) = 1, \quad \left(\frac{2}{q}\right) = 1, \quad \left(\frac{5}{q}\right) = -1.$$

The Legendre symbol $\left(\frac{5}{q}\right)$ is used only in the PC2 channel. It functions as a 5-wheel orientation gate.

5 Why the joint tile is $80 = 2^4 \cdot 5$

The strong selector does not select from an incomplete prefix segment. It begins at the first completed joint transport tile. Two carrier structures must be jointly visible: the 2-adic lane carrier and the PC2 5-wheel carrier.

Why the 2-adic carrier is 16, not 8

The deepest declared 2-adic branch in the channel grammar is the α lane $v_2(w - 1) = 3$. To test equality $v_2(w - 1) = 3$ rather than merely $v_2(w - 1) \geq 3$, the carrier must distinguish divisibility by 2^3 from divisibility by 2^4 . This requires visibility modulo $2^{3+1} = 16$. Modulo 8 can see that $w - 1$ is divisible by 8, but it cannot distinguish the exact branch $v_2 = 3$ from deeper branches $v_2 \geq 4$. Therefore the exact lane carrier for the deepest $v_2 = 3$ branch is $\mathbb{Z}/16\mathbb{Z}$.

Why the 5-wheel is included

The PC2 channel uses the wheel orientation $\left(\frac{5}{q}\right) = -1$. The corresponding wheel carrier is the mod 5 component. Although α and SU2 do not individually use the $\left(\frac{5}{q}\right)$ gate, TPCS is a triple selector, not three unrelated selectors stitched together. The output is selected on a shared transport tile carrying every coordinate needed by the joint mechanism.

The shared tile must therefore carry both $\mathbb{Z}/16\mathbb{Z}$ and $\mathbb{Z}/5\mathbb{Z}$. By CRT,

$$\mathbb{Z}/16\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \cong \mathbb{Z}/80\mathbb{Z}.$$

The scalar form of the joint tile is $M = 80$. The first completed joint tile is $80 \leq w \leq 159$.

This is not merely the lower bound where Ω appears. It is the first completed tile of the joint coordinate system required by the declared triple mechanism.

6 The strong joint-tile selector

Definition 2 (Canonical strong TPCS selector). For each channel $C \in \{\alpha, SU2, PC2\}$, select the typed carrier pair (w, q) with minimal displayed value w , subject to:

1. $w \geq 80$,

2. w is prime,
3. q is an odd prime divisor of $w - 1$,
4. $q \equiv 1 \pmod{4}$,
5. $q > \sqrt{w}$,
6. the channel-specific Legendre and v_2 conditions hold.

The output of the channel is the selected typed pair. The displayed channel value is the w -projection.

The readout policy is therefore: minimize w . The carrier q certifies the typed mechanism; it is not the displayed value being minimized. Alternative policies such as minimizing $w + q$, minimizing q , or maximizing q define different selector mechanisms.

7 Main theorems

Theorem 3 (Prefix obstruction). *For $3 \leq w < 80$, the canonical TPCS gates produce:*

$$\alpha : \emptyset, \quad \text{SU2} : (11, 5), (59, 29), (79, 13), \quad \text{PC2} : \emptyset.$$

Proof. The finite prefix table in Appendix B exhausts every prime $w < 80$, every odd prime divisor $q \mid (w - 1)$, and every channel gate. The only passing typed carriers are $(11, 5), (59, 29), (79, 13)$ in the SU2 channel. No α or PC2 carrier passes in the prefix regime. $\square$

Remark 4. The prefix regime does not contain all three channel mechanisms. It contains only SU2 prefix carriers. The unrestricted statement “TPCS selects Ω over all positive primes” is therefore false. The correct statement is: TPCS selects Ω in the first completed joint transport tile.

Theorem 5 (Canonical TPCS finite minimality). *Under the canonical strong joint-tile TPCS rules, the first typed carrier pairs are*

$$\alpha = (137, 17), \quad \text{SU2} = (107, 53), \quad \text{PC2} = (103, 17).$$

Proof. It suffices to prove three finite minimum claims. For α , because the proposed minimum is $w = 137$, check prime w in $80 \leq w < 137$. For SU2, because the proposed minimum is $w = 107$, check prime w in $80 \leq w < 107$. For PC2, because the proposed minimum is $w = 103$, check prime w in $80 \leq w < 103$.

For each prime w in the relevant interval, every odd prime divisor $q \mid (w - 1)$ is tested against the TPCS gates. The target pair is then verified to pass.

The finite proof tables are given in Appendix A. They show:

1. no α carrier exists for $80 \leq w < 137$,
2. no SU2 carrier exists for $80 \leq w < 107$,
3. no PC2 carrier exists for $80 \leq w < 103$,
4. $(137, 17), (107, 53), (103, 17)$ pass all gates.

Hence the three selected typed pairs are exactly $(137, 17), (107, 53), (103, 17)$. $\square$

Theorem 6 (Projected Ω). *Under the canonical strong joint-tile TPCS selector, the typed carrier triple $((137, 17), (107, 53), (103, 17))$ has w -projection*

$$\Omega = (137, 107, 103).$$

Proof. Apply $\pi_w(w, q) = w$ componentwise to Theorem 5. □

8 Selector hierarchy compression

Theorem 7 (Selector hierarchy compression). *Let TPCS gates be added in the canonical order:*

1. *prime w ,*
2. *typed carrier $q \mid (w - 1)$,*
3. *$q \equiv 1 \pmod{4}$,*
4. *$q > \sqrt{w}$,*
5. *Legendre $\left(\frac{2}{q}\right)$ channel orientation,*
6. *$v_2(w - 1)$ lane,*
7. *PC2 5-wheel,*
8. *first completed joint tile.*

Then the projected first triples evolve as

$$\begin{aligned} (3, 3, 3) &\rightarrow (7, 7, 7) \rightarrow (11, 11, 11) \rightarrow (11, 11, 11) \rightarrow \\ (83, 11, 83) &\rightarrow (137, 11, 83) \rightarrow (137, 11, 103) \rightarrow (137, 107, 103). \end{aligned}$$

Removing the repeated no-change stage gives the compression path

$$(3, 3, 3) \rightarrow (7, 7, 7) \rightarrow (11, 11, 11) \rightarrow (83, 11, 83) \rightarrow (137, 11, 83) \rightarrow (137, 11, 103) \rightarrow (137, 107, 103).$$

Proof. At each gate level, compute the channel-wise minimal w -projection of admissible typed carriers. Each transition is caused by the newly imposed gate: typed carrier removes prime-only rows; $q \equiv 1 \pmod{4}$ removes $q \equiv 3 \pmod{4}$ carriers; Legendre $\left(\frac{2}{q}\right)$ splits channel orientation; v_2 -lane selects the exact 2-adic branch; PC2 5-wheel orients PC2; joint-tile boundary removes prefix SU2 carriers. □

9 Causal component entry

Theorem 8 (Causal component entry). *The coordinates of Ω enter through three distinct causal gates:*

$$\begin{aligned} 137 &\leftarrow v_2 = 3 \text{ (}\alpha \text{ lane),} \\ 103 &\leftarrow \text{PC2 5-wheel,} \\ 107 &\leftarrow \text{first completed joint-tile reset.} \end{aligned}$$

Proof. Before the v_2 -lane gate, the first α -like carrier is $(83, 41)$. But $83 - 1 = 82 = 2 \cdot 41$, so $v_2(83 - 1) = 1 \neq 3$. Hence $(83, 41)$ is eliminated by the α lane, and the next first α carrier is $(137, 17)$.

Before the PC2 5-wheel gate, the first PC2-like carrier is also $(83, 41)$. It fails the wheel because $\left(\frac{5}{41}\right) = 1 \neq -1$. The next first PC2 carrier is $(103, 17)$.

Before the joint-tile reset, the SU2 prefix carriers are $(11, 5), (59, 29), (79, 13)$. These are excluded when the selector enters the first completed joint tile $w \geq 80$. The next first SU2 carrier is $(107, 53)$. $\square$

10 Mechanistic reset

The selector has two regimes.

Prefix regime

The prefix segment is $3 \leq w < 80$. It is incomplete modulo 80, covering only 77/80 scalar mod-80 residues. It contains exactly three typed carriers, all in the SU2 channel:

$$(11, 5), \quad (59, 29), \quad (79, 13).$$

No α carrier and no PC2 carrier exists in this prefix regime.

First completed joint-tile regime

The first completed joint tile is $80 \leq w \leq 159$. In this tile, the first channel-wise typed carriers are $\alpha = (137, 17)$, $SU2 = (107, 53)$, $PC2 = (103, 17)$.

The reset is therefore

$$\text{prefix-only SU2 regime} \longrightarrow \text{completed joint three-channel regime}.$$

This is the mechanistic meaning of $M = 80$.

11 Negative tile controls

A natural challenge: why not $M = 40, 72, 96, 100$, or 160? A negative tile control audit was conducted on natural candidate tile moduli. Only $M = 80$ survived all declared joint transport requirements.

Smaller candidates

- $M = 40$ lacks the required mod 16 lane carrier and retains the lower SU2 obstruction.
- $M = 72$ lacks both the mod 16 lane carrier and mod 5 wheel carrier as a joint mod 80 tile, and yields $(137, 79, 103)$.
- $M = 75$ also yields $(137, 79, 103)$, but lacks the mod 16 lane carrier and is not a multiple of the joint mod 80 structure.

Near-miss candidates

Some candidates, such as $M = 96$ and $M = 100$, reproduce the projected Ω under the same lower-bound readout, but fail the structural tile test.

- $M = 96$ lacks the mod 5 wheel carrier and is not a multiple of the joint mod 80 structure.
- $M = 100$ lacks the mod 16 lane carrier and is not a multiple of the joint mod 80 structure.

They work as numerical lower bounds, but not as valid joint transport tiles. This separates two notions: *numerically reproduces Ω* versus *is a valid joint transport tile*. TPCS uses the second criterion.

Overshoot candidates

$M = 160$ is a multiple of 80 but skips the first completed Ω tile. It yields $(1097, 223, 227)$. Later complete joint tiles are not equivalent to the first completed joint tile.

Conclusion

Among the natural control set, $M = 80$ is the first candidate that carries the mod 16 lane, carries the mod 5 wheel, carries the joint mod 80 structure, contains first completed three-channel coexistence, and produces Ω . This is why $M = 80$ is structurally distinguished from $M = 81, 88, 96, 100, 160$. It is not distinguished merely by the fact that it produces Ω .

12 Readout policy invariance

The TPCS output is the displayed channel value $\pi_w(w, q) = w$. The admissible readout policy is therefore: channel-wise minimal w . Under all tested readout-compatible policies, the typed outputs are invariant:

$$(137, 17), \quad (107, 53), \quad (103, 17).$$

Alternative policies such as minimizing $w + q$, minimizing q , or maximizing q define different mechanisms. For example, minimizing $w + q$ changes the SU2 tile-local choice from $(107, 53)$ to $(131, 13)$, because $107 + 53 = 160$ while $131 + 13 = 144$. That is not a failure of TPCS; it is a different selector objective.

13 Counterfactual teeth

The mechanism has counterfactual teeth: removing or perturbing layers degrades the output predictably.

Remove typed carrier: the completed-tile output becomes $(83, 83, 83)$.

Remove Legendre $(\frac{2}{q})$: output becomes $(137, 83, 103)$.

Remove v_2 -lane: output becomes $(83, 107, 103)$.

Remove PC2 5-wheel: output becomes $(137, 107, 83)$.

Remove joint tile: output becomes $(137, 11, 103)$.

Removing structural layers does not produce random noise; it produces interpretable mechanism failure.

14 Structural hygiene roles

Not every gate is first-output critical. The audits classify gates into two main roles:

1. *Selection-critical gates*: removing them changes the projected first triple.
2. *Structural-hygiene gates*: removing them may preserve the first triple but admits off-mechanism rows.

The gates $q \equiv 1 \pmod{4}$ and $q > \sqrt{w}$ are structural-hygiene gates in the tested range. Removing them does not change the first output, but it admits rows that violate orientation or extremality. For example, removing $q \equiv 1 \pmod{4}$ admitted 326 wrong-orientation rows; removing $q > \sqrt{w}$ admitted 603 non-extremal carrier rows.

The correct claim is therefore not that every gate is first-output critical, but that each gate has a typed role: selection, orientation, extremality, lane, wheel, reset, or structural hygiene.

15 Core kernel isolation

The selector separates into a causal core and a hygiene envelope.

Core kernel

$$\alpha : v_2 = 3, \left(\frac{2}{q}\right) = 1, \quad SU2 : v_2 = 1, \left(\frac{2}{q}\right) = -1, \quad PC2 : v_2 = 1, \left(\frac{2}{q}\right) = 1.$$

Hygiene envelope

$$q \equiv 1 \pmod{4}, \quad q > \sqrt{w}, \quad \left(\frac{5}{q}\right) = -1 \text{ for PC2}, \quad M = 80.$$

With the hygiene envelope fixed and the core varied across 288 nearby kernels, the exact typed Ω output occurred exactly once: 1/288. The unique successful kernel was the canonical one. By contrast, fixing the core and varying the hygiene envelope preserved typed Ω in 24/168 cases.

The core kernel is the causal source of the typed triple, while the hygiene envelope makes the mechanism canonical, interpretable, and transport-aligned.

16 Adversarial grammar neighborhood

A broader adversarial grammar neighborhood of 55,296 nearby variants was tested. The exact typed Ω output appeared in 24 cases:

$$24/55,296 \approx 4.34 \times 10^{-4}.$$

Ω is therefore rare but not globally unique over a broad grammar neighborhood. The successful nearby variants preserve the same core kernel

$$\alpha(v_2 = 3, \ell_2 = 1), \quad SU2(v_2 = 1, \ell_2 = -1), \quad PC2(v_2 = 1, \ell_2 = 1).$$

The variations occur mostly in hygiene and boundary choices, such as $q \equiv 1 \pmod{4}$ on or off, $q > \sqrt{w}$ on or off, nearby lower bounds $M \in \{81, 88, 96, 100\}$, and the PC2 5-wheel omitted in some cases.

This audit prevents overclaiming. The paper does not claim global uniqueness over all grammars. It claims a finite theorem for a defined mechanism and local robustness around a narrow typed kernel.

17 Independent proof-kernel crosscheck

The finite theorem was independently checked by two proof kernels.

Engine A. Used trial-division primality, factorization-based candidate extraction, and `pow`-based Legendre calculation.

Engine B. Used sieve-generated prime table, direct divisor enumeration, and explicit residue-square Legendre calculation.

Both engines agreed exactly on: primality and factorization for all integers $2 \leq n < 160$; prefix obstruction rows; target pass rows; no-earlier minimality intervals; full finite failure tables. The crosscheck compared 158 primality/factorization rows; 3 prefix channel tables; 3 target pass checks; 3 no-earlier minimality checks; and 19 finite failure-table rows across α , SU2, and PC2. The independent crosscheck returned hard failures = 0.

18 Proof-object mutation verifier

The finite theorem was converted into an explicit proof object. An independent verifier recomputed the proof object and checked: $M = 80$; expected typed pairs; prefix table; target pass table; no-earlier tables; proof-object SHA-256.

Then ten adversarial mutations were applied: mutate M to 81; mutate expected α ; remove SU2 prefix obstruction (79, 13); add fake α prefix row; corrupt PC2 target q ; mark $\alpha w = 83$ as passing; delete $\alpha w = 83$ no-earlier row; corrupt SU2 failure reason; remove PC2 target candidate 17; tamper with the proof hash. The verifier rejected all ten (10/10). The proof object has adversarial teeth.

19 Summary of results

Proved

1. the canonical TPCS strong joint-tile finite minimum theorem;
2. the prefix obstruction theorem;
3. the projected Ω theorem;
4. independent proof-kernel agreement;
5. proof-object verifier integrity;
6. causal mechanism decomposition;
7. policy invariance for w -minimal readout;
8. negative tile control behavior;
9. counterfactual teeth;
10. core-kernel isolation in the tested neighborhood.

Not proved

1. that nature must use TPCS;
2. that TPCS is the only possible selector grammar;
3. that Ω is globally unique over all conceivable grammars;
4. that every gate is first-output critical.

20 Conclusion

The strong joint-tile member of the TPCS family acts on typed carrier pairs (w, q) , selects channel-wise minimal w -values after the first completed joint transport tile, and produces the typed carrier triple

$$((137, 17), (107, 53), (103, 17)).$$

Projecting to w -coordinates yields $\Omega = (137, 107, 103)$.

The theorem is finite. It does not require a long search; it requires only the finite no-earlier intervals

$$80 \leq w < 137, \quad 80 \leq w < 107, \quad 80 \leq w < 103.$$

The surrounding audits show that the result is not merely a one-line fit. The selector has a family hierarchy, causal eliminations, a mechanistic reset, negative tile controls, policy invariance, counterfactual teeth, core-kernel isolation, independent proof-kernel validation, and proof-object mutation resistance.

The honest final statement is: TPCS does not prove that Ω is globally inevitable. It proves that a precisely stated finite selector mechanism selects Ω as its first completed joint-tile output, and that this result is independently certifiable.

A Finite no-earlier tables**A.1 α : $80 \leq w < 137$. Target $(137, 17)$.**

w	factorization of $w - 1$	candidate q	failure reason
83	$2 \cdot 41$	41	$v_2 = 1 \neq 3$
89	$2^3 \cdot 11$	11	$q \not\equiv 1 \pmod{4}$, wrong $(2/q)$
97	$2^5 \cdot 3$	3	$q \not\equiv 1 \pmod{4}$, $q \leq \sqrt{w}$, wrong $(2/q)$, $v_2 = 5 \neq 3$
101	$2^2 \cdot 5^2$	5	$q \leq \sqrt{w}$, wrong $(2/q)$, $v_2 = 2 \neq 3$
103	$2 \cdot 3 \cdot 17$	3	$q \not\equiv 1 \pmod{4}$, $q \leq \sqrt{w}$, wrong $(2/q)$, $v_2 = 1 \neq 3$
103	$2 \cdot 3 \cdot 17$	17	$v_2 = 1 \neq 3$
107	$2 \cdot 53$	53	wrong $(2/q)$, $v_2 = 1 \neq 3$
109	$2^2 \cdot 3^3$	3	$q \not\equiv 1 \pmod{4}$, $q \leq \sqrt{w}$, wrong $(2/q)$, $v_2 = 2 \neq 3$
113	$2^4 \cdot 7$	7	$q \not\equiv 1 \pmod{4}$, $q \leq \sqrt{w}$, $v_2 = 4 \neq 3$
127	$2 \cdot 3^2 \cdot 7$	3	$q \not\equiv 1 \pmod{4}$, $q \leq \sqrt{w}$, wrong $(2/q)$, $v_2 = 1 \neq 3$
127	$2 \cdot 3^2 \cdot 7$	7	$q \not\equiv 1 \pmod{4}$, $q \leq \sqrt{w}$, $v_2 = 1 \neq 3$
131	$2 \cdot 5 \cdot 13$	5	$q \leq \sqrt{w}$, wrong $(2/q)$, $v_2 = 1 \neq 3$
131	$2 \cdot 5 \cdot 13$	13	wrong $(2/q)$, $v_2 = 1 \neq 3$

No earlier α carrier exists, and $(137, 17)$ is first.

A.2 SU2: $80 \leq w < 107$. **Target** $(107, 53)$.

w	factorization of $w - 1$	candidate q	failure reason
83	$2 \cdot 41$	41	wrong $(2/q)$: got 1, expected -1
89	$2^3 \cdot 11$	11	$q \not\equiv 1 \pmod{4}$, $v_2 = 3 \neq 1$
97	$2^5 \cdot 3$	3	$q \not\equiv 1 \pmod{4}$, $q \leq \sqrt{w}$, $v_2 = 5 \neq 1$
101	$2^2 \cdot 5^2$	5	$q \leq \sqrt{w}$, $v_2 = 2 \neq 1$
103	$2 \cdot 3 \cdot 17$	3	$q \not\equiv 1 \pmod{4}$, $q \leq \sqrt{w}$
103	$2 \cdot 3 \cdot 17$	17	wrong $(2/q)$: got 1, expected -1

No earlier SU2 carrier exists, and $(107, 53)$ is first.

A.3 PC2: $80 \leq w < 103$. **Target** $(103, 17)$.

w	factorization of $w - 1$	candidate q	failure reason
83	$2 \cdot 41$	41	wrong $(5/q)$: got 1, expected -1
89	$2^3 \cdot 11$	11	$q \not\equiv 1 \pmod{4}$, wrong $(2/q)$, $v_2 = 3 \neq 1$, wrong $(5/q)$
97	$2^5 \cdot 3$	3	$q \not\equiv 1 \pmod{4}$, $q \leq \sqrt{w}$, wrong $(2/q)$, $v_2 = 5 \neq 1$
101	$2^2 \cdot 5^2$	5	$q \leq \sqrt{w}$, wrong $(2/q)$, $v_2 = 2 \neq 1$, $(5/q) = 0 \neq -1$

No earlier PC2 carrier exists, and $(103, 17)$ is first.

B Prefix regime

For $3 \leq w < 80$:

Channel	Prefix carriers
α	none
SU2	$(11, 5)$, $(59, 29)$, $(79, 13)$
PC2	none

This is the prefix obstruction. It is why the global all-prime claim is false and why the completed joint-tile theorem is the correct formulation.

C Canonical strong TPCS definition

The canonical strong selector is $\text{TPCS}_{80}^{\text{strong}}$. It acts on pairs (w, q) with

$$\begin{aligned}
&w \geq 80, \quad w \text{ prime}, \\
&q \text{ odd prime}, \quad q \mid (w - 1), \\
&q \equiv 1 \pmod{4}, \quad q > \sqrt{w}.
\end{aligned}$$

Channel gates:

$$\begin{aligned}
\alpha : & \quad v_2(w - 1) = 3, \quad \left(\frac{2}{q}\right) = 1, \\
SU2 : & \quad v_2(w - 1) = 1, \quad \left(\frac{2}{q}\right) = -1, \\
PC2 : & \quad v_2(w - 1) = 1, \quad \left(\frac{2}{q}\right) = 1, \quad \left(\frac{5}{q}\right) = -1.
\end{aligned}$$

Readout: $(w, q) \mapsto w$. Selection: choose channel-wise minimal w . Output:

$$((137, 17), (107, 53), (103, 17)), \quad \Omega = (137, 107, 103).$$

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